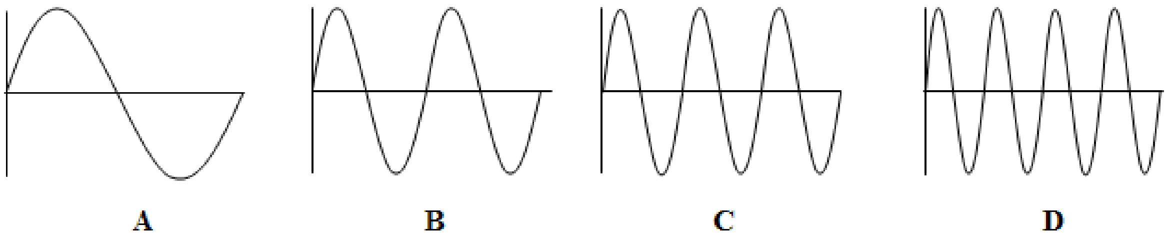
PHYS121 Integrated Science-Physics

W4 Chapter 17 Sound

References:
[1] David Halliday, Jearl Walker, Resnick Jearl, ‘Fundamentals of Physics’, (Wiley, 2018)
[2] Doug Giancoli, ‘Physics for Scientists and Engineers with modern physics’, (Pearson, 2009)
[3] Hugh D. Young, Roger A. Freedman, ‘University Physics with Modern Physics’, (Pearson, 2012)
And others specified when needed.

昆山杜克大学
DUKE KUNSHAN UNIVERSITY

16.6.2. Two identical strings each have one end attached to a wall. The other ends are each attached to a separate spool that allows the tension of each string to be changed independently. Consider each of the waves shown. Which one of the following statements is true *if the frequency and amplitude of the waves is the same*?



a) The tension in the string on which wave A is traveling is four times that in the string on which wave D is traveling.

b) The tension in the string on which wave B is traveling is four times that in the string on which wave D is traveling.

c) The tension in the string on which wave B is traveling is four times that in the string on which wave A is traveling.

d) The tension in the string on which wave D is traveling is four times that in the string on which wave A is traveling.

e) The tension in the string on which wave C is traveling is four times that in the string on which wave B is traveling.

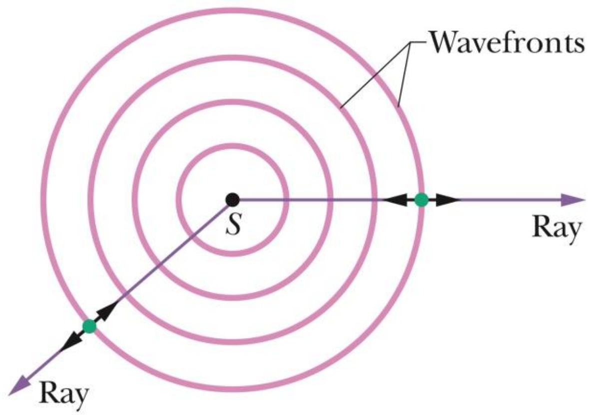
Learning Outcomes

- Compare the speeds of sound in various kinds of medium.
- Explain the wave function of sound (displacement/pressure change).
- Analyze the characteristics and behaviors of sound wave (longitudinal), i.e., amplitude, interference, standing wave (pipe), beat, etc.
- Find the intensity of sound and sound level (dB).
- Explain/Calculate the Doppler Effect of sound.
- Describe shock waves and application of ultrasound imaging.

Speed of Sound

Sound waves are longitudinal mechanical waves that can travel through solids, liquids, or gases.

Point *S* represents a tiny sound source, called a **point source**, that emits sound waves in all directions. A sound wave travels from a point source *S* through a three-dimensional medium. The **wavefronts (**surfaces over which the oscillations due to the sound wave have the same value) form spheres centered on *S*; the **rays** are radial to *S*. The short, double-headed arrows indicate that elements of the medium oscillate parallel to the rays.

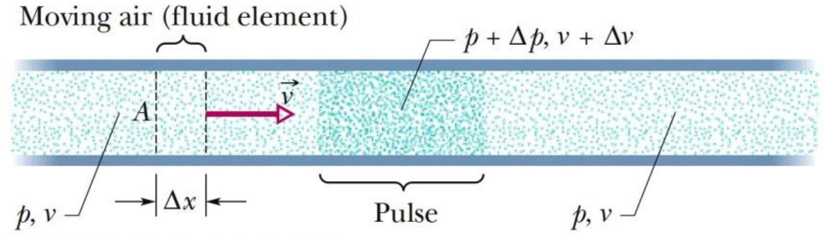


$$F = pA - (p + \Delta p)\, A = -\Delta p\, A \quad \text{(net force).}$$

$$\Delta m = \rho\, \Delta V = \rho A\, \Delta x = \rho A v\, \Delta t \quad \text{(mass).}$$

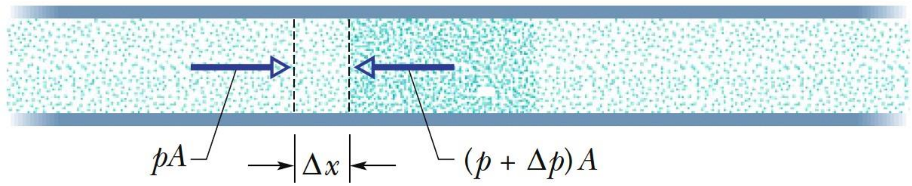
The speed v of a sound wave in a medium having bulk modulus B and density ρ is

$$v = \sqrt{\frac{B}{\rho}} \quad \text{(speed of sound)}$$



Through direct application of Newton's Second law.

An element of air of width Δx moves toward the pulse with speed v.



$$B = -\frac{\Delta p}{\Delta V/V} \quad \text{(definition of bulk modulus).}$$

The leading face of the element enters the pulse. The forces acting on the leading and trailing faces (due to air pressure) are shown.

Sound can travel through any kind of matter, but not through a vacuum (check).

TABLE 16–1 Speed of Sound in Various Materials (20°C and 1 atm)

Material	Speed (m/s)
Air	343
Air (0°C)	331
Helium	1005
Hydrogen	1300
Water	1440
Sea water	1560
Iron and steel	≈ 5000
Glass	≈ 4500
Aluminum	≈ 5100
Hardwood	≈ 4000
Concrete	≈ 3000

The speed of sound is different in different materials; in general, it is slowest in gases, faster in liquids, and fastest in solids.

The speed depends somewhat on temperature, especially for gases.

Loudness: related to **intensity** of the sound wave

Pitch: related to **frequency**

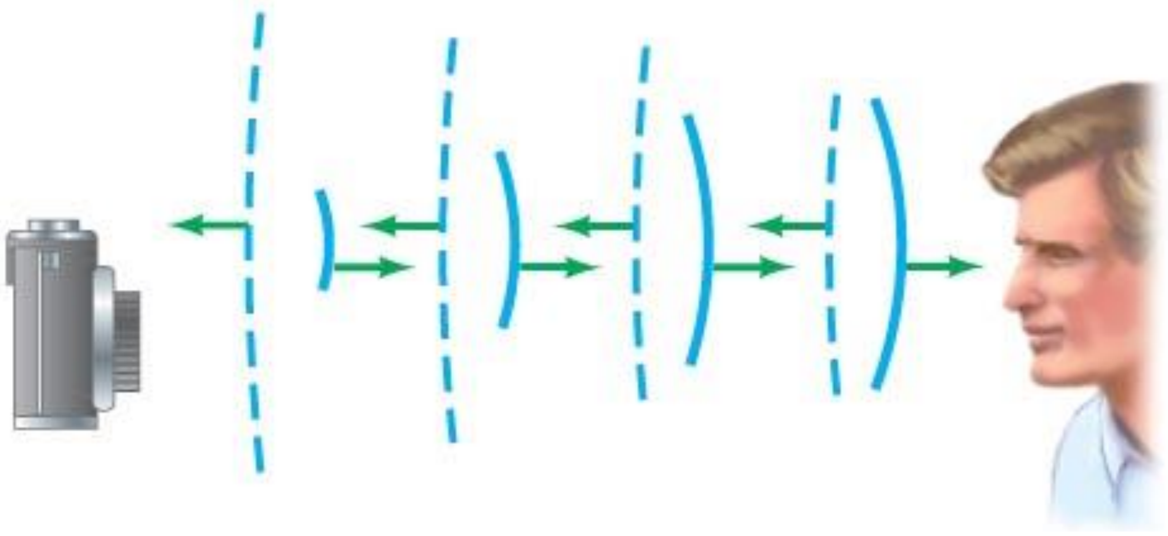
Audible range: about **20** Hz to **20,000** Hz; upper limit decreases with age

Ultrasound: above **20,000** Hz; see ultrasonic camera focusing in following example

Infrasound: below **20** Hz

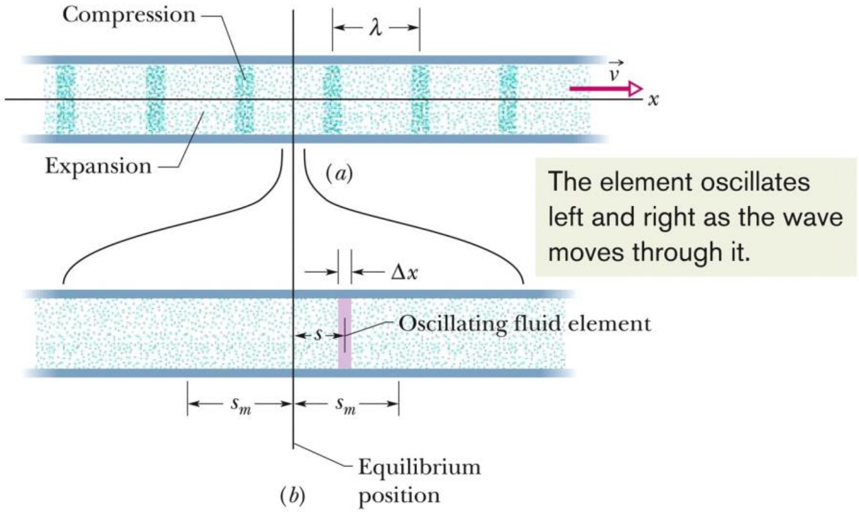
Example: Autofocusing with sound waves.

Older autofocusing cameras determine the distance by emitting a pulse of very high frequency (ultrasonic) sound that travels to the object being photographed, and include a sensor that detects the returning reflected sound. To get an idea of the time sensitivity of the detector, calculate the travel time of the pulse for an object (a) 1.0 m away, and (b) 20 m away.



Traveling Sound Waves

(a) A sound wave, traveling through a long air-filled tube with speed v, consists of a moving, periodic pattern of expansions and compressions of the air. The wave is shown at an arbitrary instant.



(b) A horizontally expanded view of a short piece of the tube. As the wave passes, an air element of thickness Δx oscillates left and right in simple harmonic motion about its equilibrium position. At the instant shown in (b), the element happens to be displaced a distance s to the right of its equilibrium position. Its maximum displacement, either right or left, is s_m.

Displacement: A sound wave causes a longitudinal displacement *s* of a mass element in a medium as given by

$$s(x,t) = s_m \cos(kx - \omega t).$$

where s_m is the displacement amplitude (maximum displacement) from equilibrium, $k = \frac{2\pi}{\lambda}$, and $\omega = 2\pi f$, λ and f being the wavelength and frequency, respectively, of the sound wave.

Pressure: The sound wave also causes a pressure change Δp of the medium from the equilibrium pressure:

$$\Delta p(x,t) = \Delta p_m \sin(kx - \omega t).$$

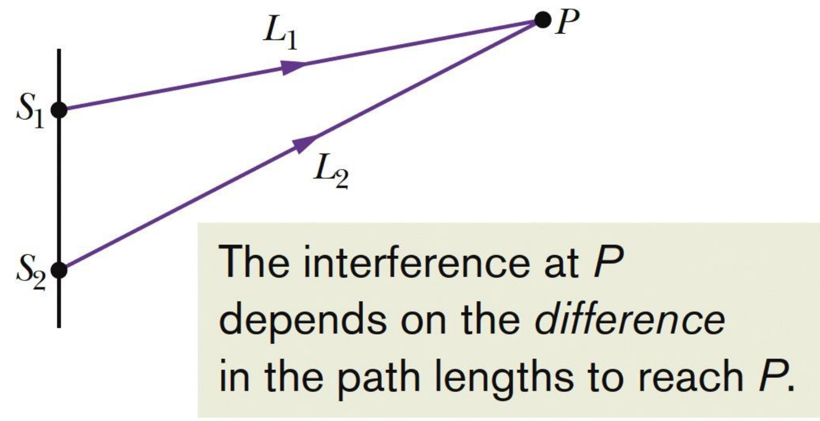
where the pressure amplitude is $\Delta p_m = (v\rho\omega) s_m$.

https://www.feynmanlectures.caltech.edu/I_47.html

Interference

Path Length Difference

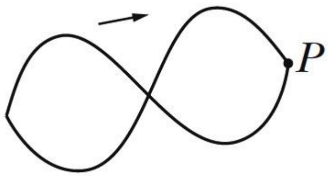
- The interference of two sound waves with *identical wavelengths* passing through a common point depends on their *phase difference*, the ϕ. If the sound waves were emitted in phase and are traveling in approximately the same direction, ϕ is given by



$$\phi = \frac{\Delta L}{\lambda} 2\pi.$$

where ΔL is their **path length difference**.

Fully Destructive Interference (exactly out of phase)

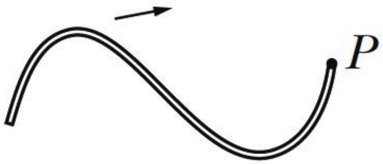


If the difference is equal to, say, 2.5λ, then the waves arrive exactly out of phase. This is how transverse waves would look.

$$\phi = (2m+1)\pi, \quad \text{for } m = 0, 1, 2, \ldots,$$

$$\frac{\Delta L}{\lambda} = 0.5, 1.5, 2.5, \ldots$$

Fully Constructive Interference (exactly in phase)



If the difference is equal to, say, 2.0λ, then the waves arrive exactly in phase. This is how transverse waves would look.

$$\phi = m(2\pi), \quad \text{for } m = 0, 1, 2, \ldots,$$

$$\frac{\Delta L}{\lambda} = 0, 1, 2, \ldots$$

Sample Problem 17.02

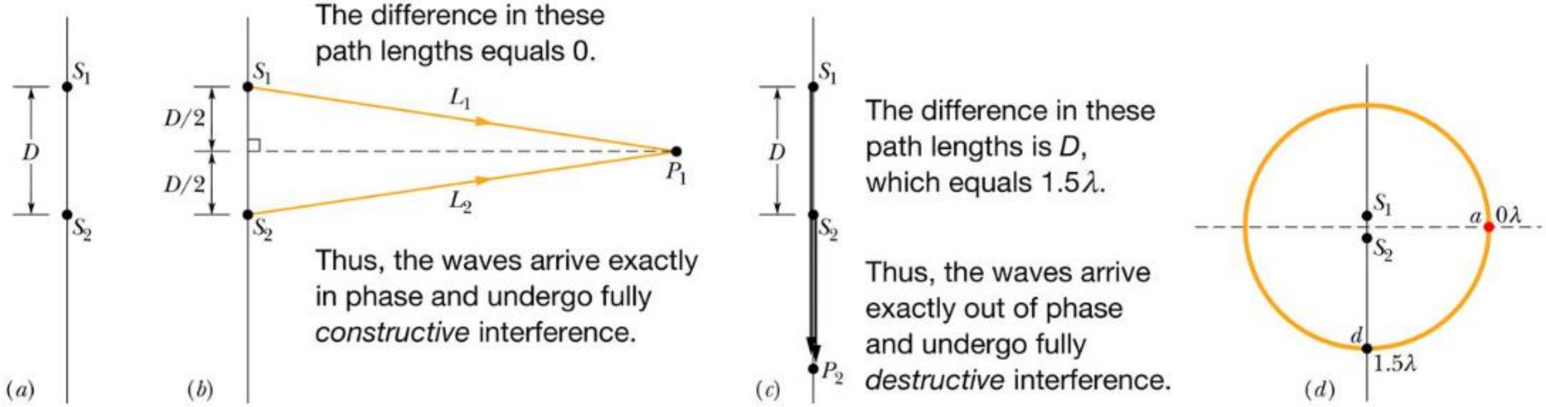
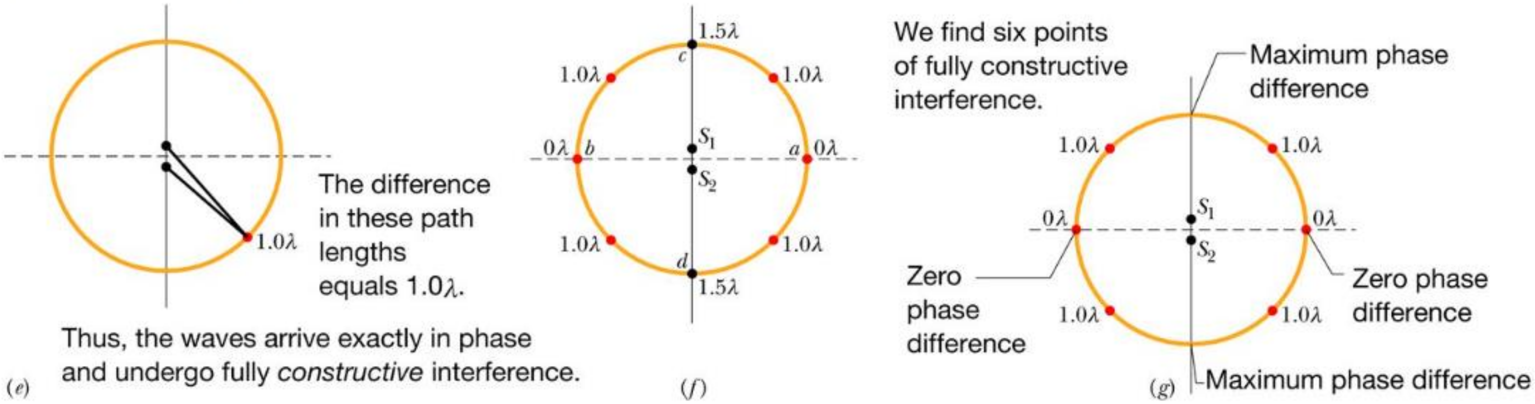


Figure 17-8 (a) Two point sources S_1 and S_2, separated by distance D, emit spherical sound waves in phase. (b) The waves travel equal distances to reach point P_1. (c) Point P_2 is on the line extending through S_1 and S_2. (d) We move around a large circle. (*Figure continues*) (e) Another point of fully constructive interference. (f) Using symmetry to determine other points. (g) The six points of fully constructive interference.



Intensity and Sound Level

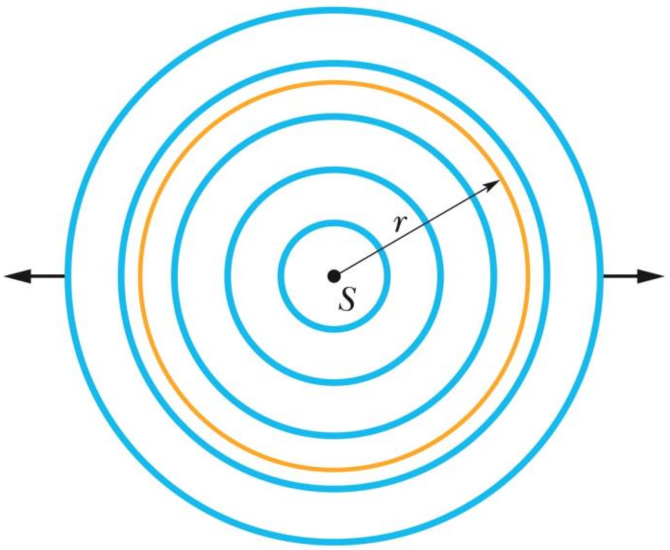
- The **intensity** I of a sound wave at a surface is the average rate per unit area at which energy is transferred by the wave through or onto the surface

$$I = \frac{P}{A},$$

where P is the time rate of energy transfer (**power**) of the sound wave and A is the area of the surface intercepting the sound. The intensity I is related to the displacement amplitude s_m of the sound wave by

$$I = \frac{1}{2}\rho v \omega^2 s_m^2.$$

- The intensity at a distance r from a point source that emits sound waves of power P_s equally in all directions isotropically i.e. with equal intensity in all directions,



$$I = \frac{P_s}{4\pi r^2},$$

where $4\pi r^2$ is the area of the sphere.

A point source S emits sound waves uniformly in all directions. The waves pass through an imaginary sphere of radius r that is centered on S.

The human ear can detect sounds with an intensity as low as 10^{-12} W/m^2 and as high as 1 W/m^2.

The Decibel Scale

- The sound level β in decibels (dB) is defined as

$$\beta = (10 \text{ dB}) \log \frac{I}{I_0}.$$

Where $I_0 (= 10^{-12} \text{ W/m}^2)$ is a reference intensity level to which all intensities are compared. For every factor-of-10 increase in intensity, 10 dB is added to the sound level.

© Ben Rose

Sound can cause the wall of a drinking glass to oscillate. If the sound produces a standing wave of oscillations and if the intensity of the sound is large enough, the glass will shatter.

TABLE 16–2
Intensity of Various Sounds

Source of the Sound	Sound Level (dB)	Intensity (W/m^2)
Jet plane at 30 m	140	100
Threshold of pain	120	1
Loud rock concert	120	1
Siren at 30 m	100	1×10^{-2}
Truck traffic	90	1×10^{-3}
Busy street traffic	80	1×10^{-4}
Noisy restaurant	70	1×10^{-5}
Talk, at 50 cm	65	3×10^{-6}
Quiet radio	40	1×10^{-8}
Whisper	30	1×10^{-9}
Rustle of leaves	10	1×10^{-11}
Threshold of hearing	0	1×10^{-12}

The intensity of a wave is the energy transported per unit time across a unit area.

Perceived loudness, however, is not proportional to the intensity.

Example 16-6: Airplane roar.

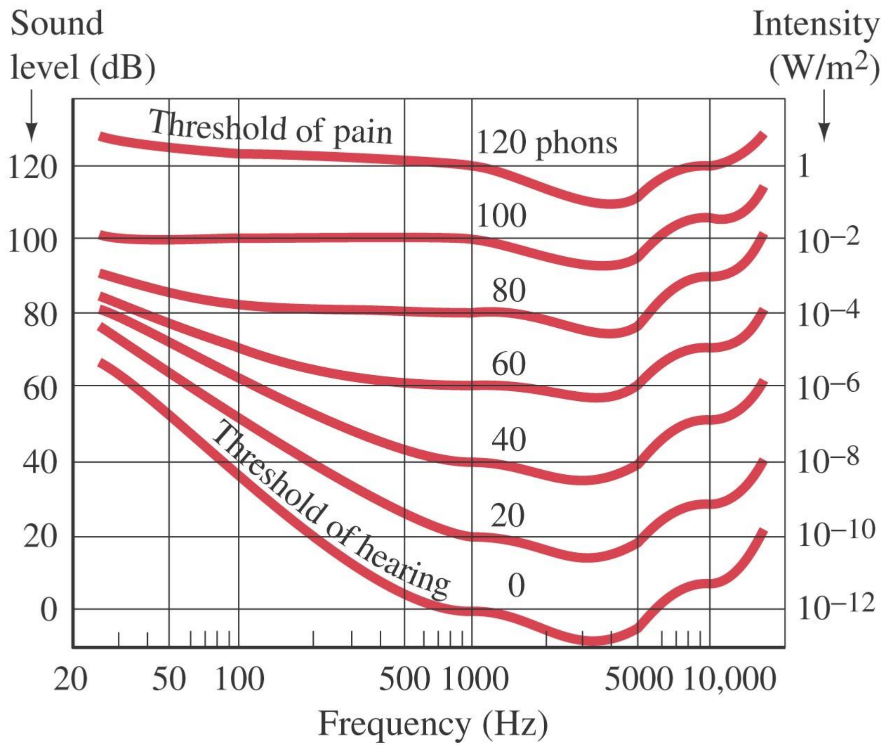
The sound level measured 30 m from a jet plane is 140 dB. What is the sound level at 300 m? (Ignore reflections from the ground.)

Example 16-7: How tiny the displacement is.

(a) Calculate the displacement of air molecules for a sound having a frequency of 1000 Hz at the threshold of hearing.

(b) Determine the maximum pressure variation in such a sound wave.

The ear's **sensitivity** varies with **frequency**. These curves translate the intensity into sound level at different frequencies.



Musical instruments produce sounds in various ways—vibrating strings, vibrating membranes, vibrating metal or wood shapes, vibrating air columns.

The vibration may be started by plucking, striking, bowing, or blowing. The vibrations are transmitted to the air and then to our ears.

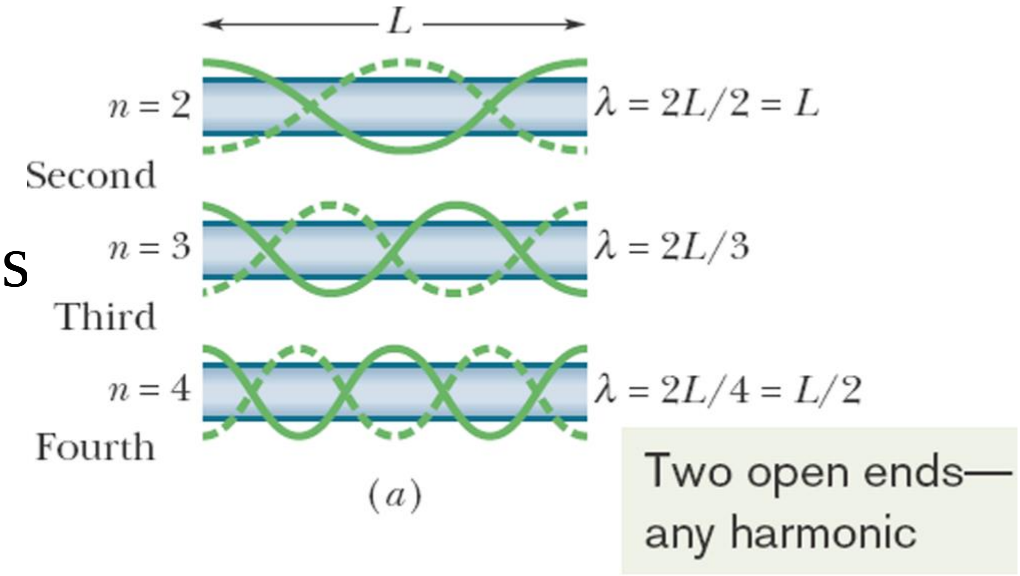
Sources of Musical Sound

Standing sound wave patterns can be set up in pipes (that is, resonance can be set up) if sound of the proper wave-length is introduced in the pipe.

Two Open Ends.

A pipe open at both ends will resonate at frequencies

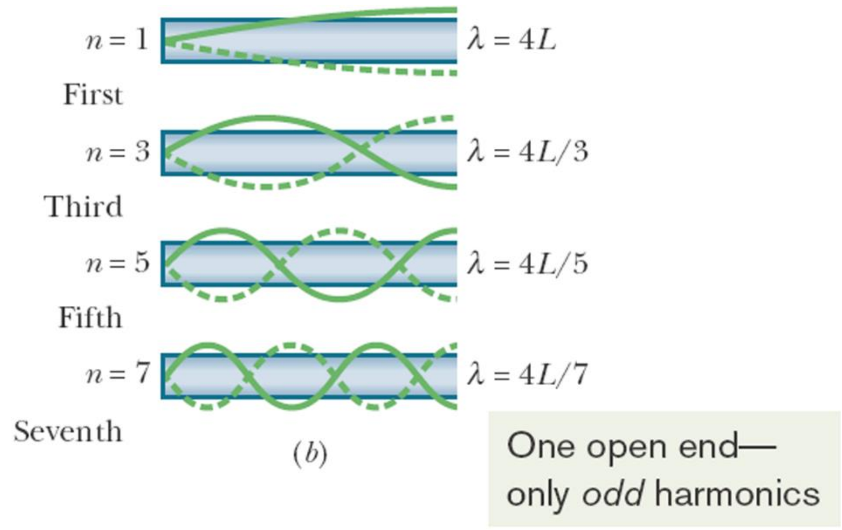
$$f = \frac{v}{\lambda} = \frac{nv}{2L}, \quad n = 1, 2, 3, \ldots,$$



One Open End.

A pipe closed at one end and open at the other will resonate at frequencies

$$f = \frac{v}{\lambda} = \frac{nv}{4L}, \quad n = 1, 3, 5, \ldots.$$



Example 16-10: Organ pipes.

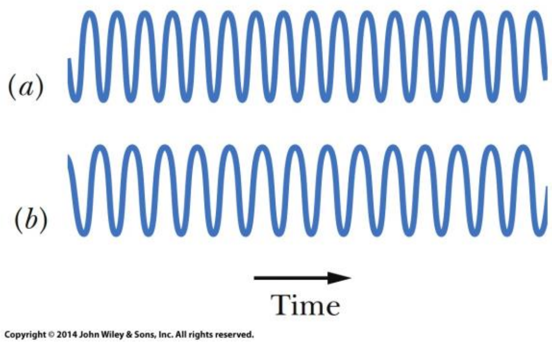
What will be the fundamental frequency and first three overtones for a 26-cm-long organ pipe at 20° C if it is (a) open and (b) closed?

Beats

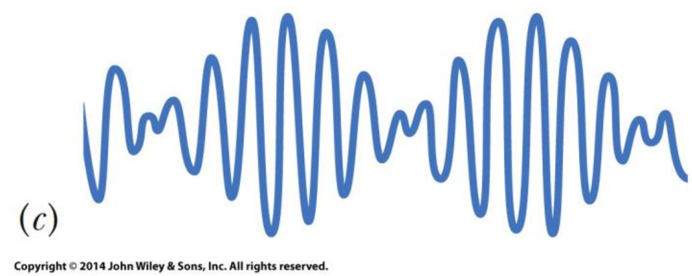
$$s = 2s_m \cos\left[\tfrac{1}{2}(\omega_1 - \omega_2)t\right] \cos\left[\tfrac{1}{2}(\omega_1 + \omega_2)\,t\right]$$

Beats arise when two waves having slightly different frequencies, f_1 and f_2, are detected together. The beat frequency (*interference*) is

$$f_{\text{beat}} = f_1 - f_2 \quad (\text{beat frequency}).$$



(a, b) The pressure variations Δp of two sound waves as they would be detected separately. The frequencies of the waves are nearly equal.



(c) The resultant pressure variation if the two waves are detected simultaneously.

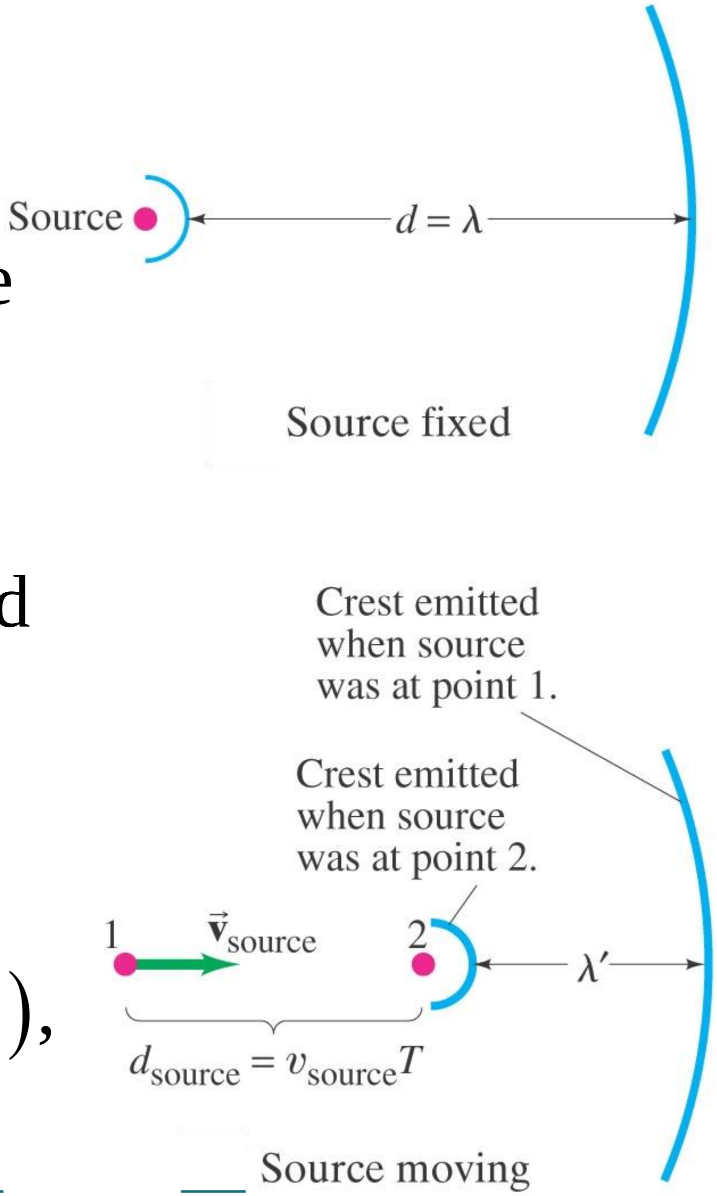
Example: Beats.

A tuning fork produces a steady 400-Hz tone. When this tuning fork is struck and held near a vibrating guitar string, twenty beats are counted in five seconds. What are the possible frequencies produced by the guitar string?

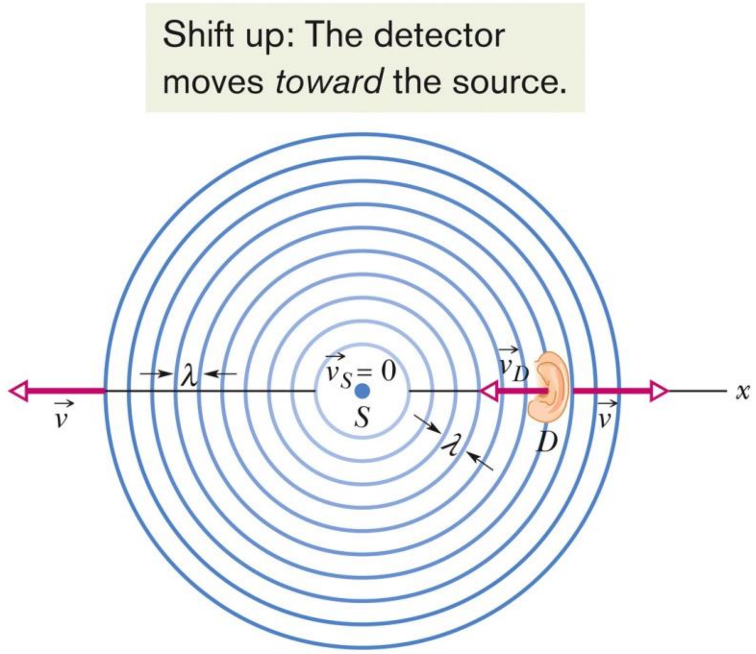
The Doppler Effect

The Doppler effect is a change in the observed frequency of a wave when the source or the detector moves *relative to the transmitting medium* (such as air). For sound, the observed frequency f' is given in terms of the source frequency f by

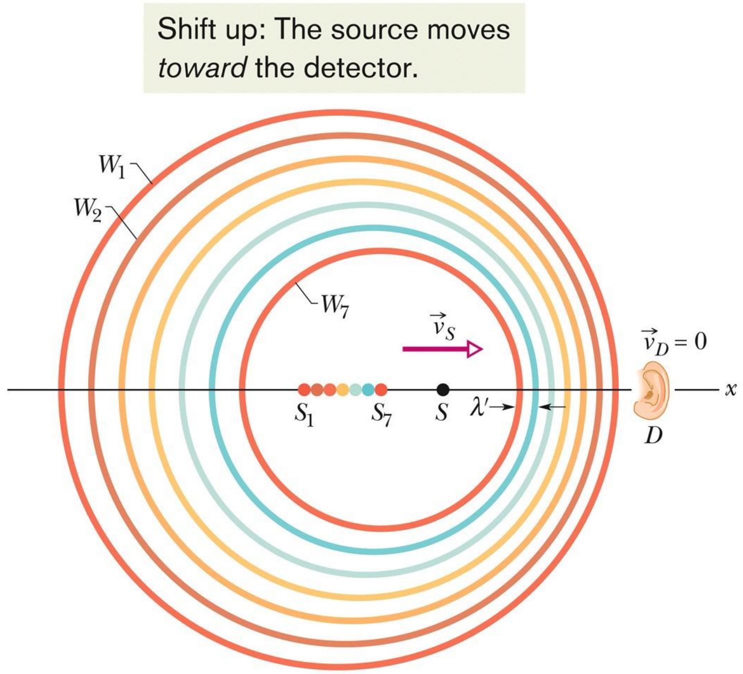
$$f' = f\,\frac{v \pm v_D}{v \pm v_S} \quad \text{(general Doppler effect)},$$



Detector Moving Source Stationary

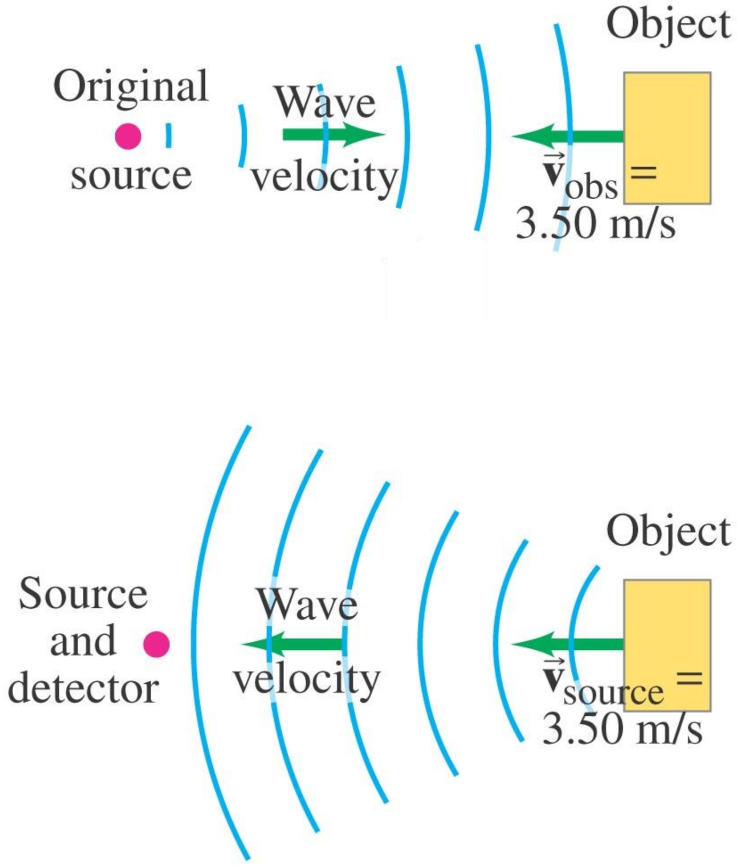


Source Moving Detector Stationary

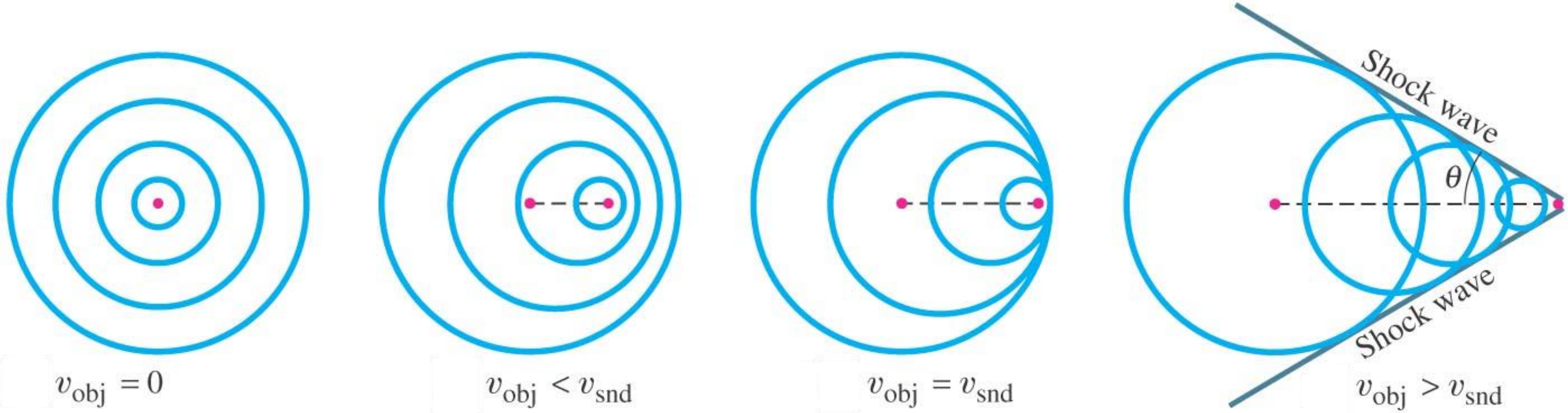


Example: Two Doppler shifts.

A 5000-Hz sound wave is emitted by a stationary source. This sound wave reflects from an object moving toward the source. What is the frequency of the wave reflected by the moving object as detected by a detector at rest near the source?



Supersonic Speeds, Shock Waves

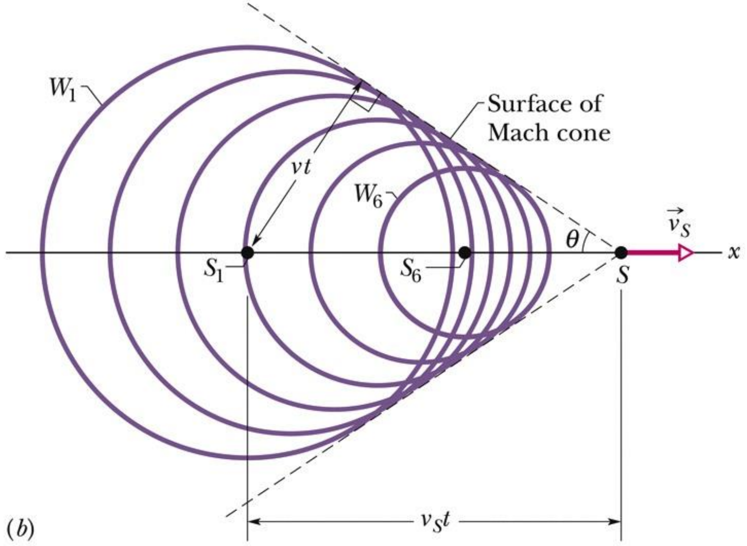


If the speed of a source relative to the medium exceeds the speed of sound in the medium, *the Doppler equation no longer applies*. In such a case, shock waves result. The half-angle θ of the Mach cone is given by

$$\sin\theta = \frac{vt}{v_s t} = \frac{v}{v_s} \quad (\text{Mach cone angle}).$$

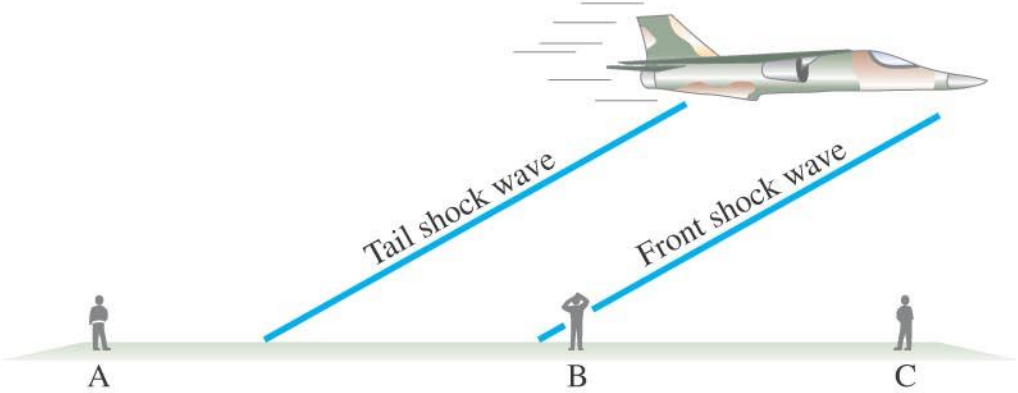
$$\sin\theta = \frac{vt}{v_S t} = \frac{v}{v_S} \quad \text{(Mach cone angle).}$$

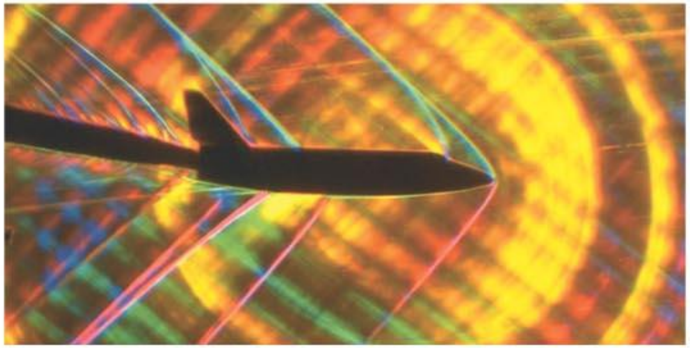
A source S moves at speed v_S faster than the speed of sound and thus faster than the wavefronts. When the source was at position S_1 it generated wavefront W_1, and at position S_6 it generated W_6. All the spherical wavefronts expand at the speed of sound v and bunch along the surface of a cone called the Mach cone, forming a shock wave. The surface of the cone has half-angle θ and is tangent to all the wavefronts.



Shock waves are analogous to the **bow waves** produced by a boat going faster than the wave speed in water.

Aircraft exceeding the speed of sound in air will produce two **sonic booms**, one from the **front** and one from the **tail**.



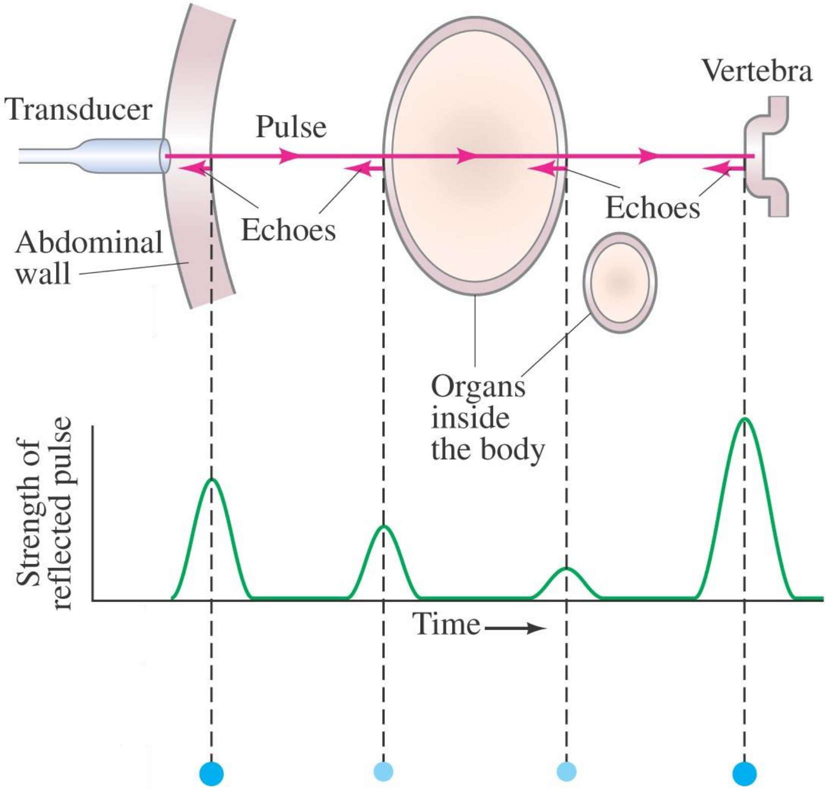


*Applications: Sonar, Ultrasound, and Medical Imaging

Sonar is used to locate objects underwater by measuring the **time** it takes a sound **pulse** to reflect back to the receiver.

Similar techniques can be used to learn about the **internal structure** of the Earth.

Sonar usually uses **ultrasound** waves, as the shorter wavelengths are less likely to be **diffracted** by obstacles.



Ultrasound is also used for **medical imaging**. Repeated traces are made as the transducer is moved, and a complete picture is built.

This is an ultrasound image of a human fetus, showing great detail.

Learning Outcomes

- ✓ Compare the speeds of sound in various kinds of medium.
- ✓ Explain the wave function of sound (displacement/pressure change).
- ✓ Analyze the characteristics and behaviors of sound wave (longitudinal), i.e., amplitude, interference, standing wave (pipe), beat, etc.
- ✓ Find the intensity of sound and sound level (dB).
- ✓ Explain/Calculate the Doppler Effect of sound.
- ✓ Describe shock waves and application of ultrasound imaging.

$$v = \sqrt{\frac{B}{\rho}}$$

$$I = \frac{1}{2}\rho v \omega^2 s_m^2.$$

$$\beta = (10\text{ dB})\log\frac{I}{I_0},$$

$$f' = f\,\frac{v \pm v_D}{v \pm v_S}$$

$$\sin\theta = \frac{vt}{v_S t} = \frac{v}{v_S} \quad (\text{Mach cone angle}).$$

昆山杜克大学 DUKE KUNSHAN UNIVERSITY

CHAPTER REVIEW AND EXAMPLES, DEMOS

https://openstax.org/books/university-physics-volume-1/pages/17-summary

$$v = 331\frac{\text{m}}{\text{s}}\sqrt{1 + \frac{T_{\text{C}}}{273°\text{C}}} = 331\frac{\text{m}}{\text{s}}\sqrt{\frac{T_{\text{K}}}{273\ \text{K}}}$$

One of the more important properties of sound is that its speed is nearly independent of the frequency. … Recall that $v = f\lambda$. In a given medium under fixed conditions, v is constant, so there is a relationship between f and λ; the higher the frequency, the smaller the wavelength.

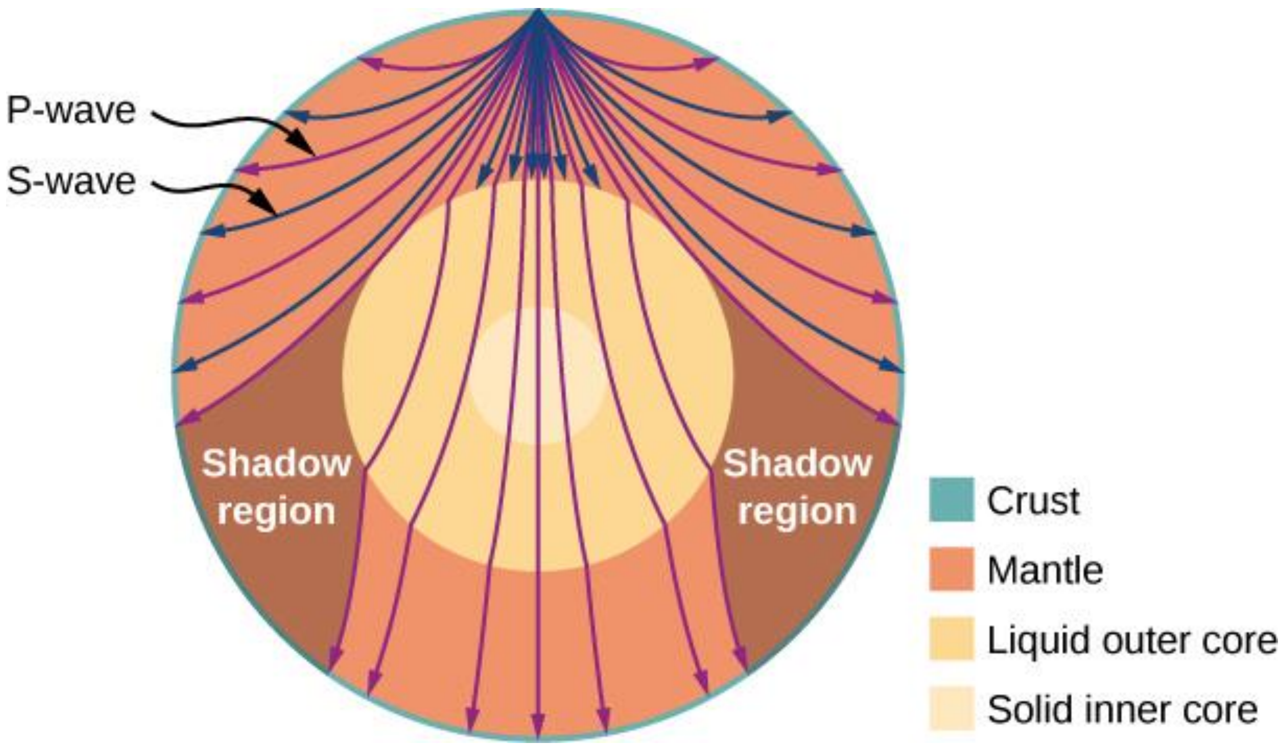
The speed of sound can change when sound travels from one medium to another, but the frequency usually remains the same.

The perception of intensity is called **loudness**. …But loudness is not related to intensity alone. Frequency has a major effect on how loud a sound seems.

$$f' = f_0 \cdot \frac{v_{snd} \pm v_w \pm v_{obs}}{v_{snd} \pm v_w \pm v_{src}}$$

https://people.highline.edu/iglozman/classes/physnotes/doppler.htm#:~:text=Wind%20has%20essentially%20the%20same,effects%20on%20the%20perceived%20frequency.

FIGURE 17.11



Earthquakes produce both longitudinal waves (P-waves) and transverse waves (S-waves), and these travel at different speeds. Both waves travel at different speeds in the different regions of Earth, but in general, P-waves travel faster than S-waves. S-waves cannot be supported by the liquid core, producing shadow regions.

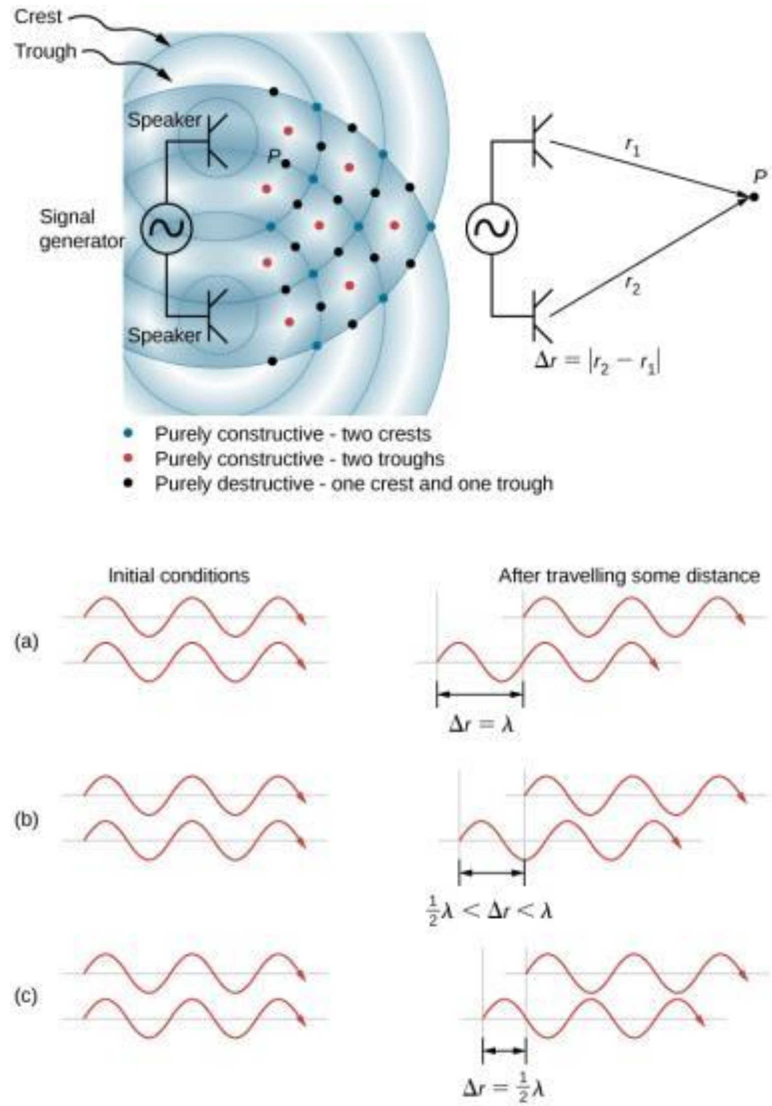
FIGURE 17.17

Two speakers being driven by a single signal generator. The sound waves produced by the speakers are in phase and are of a single frequency. The sound waves interfere with each other. When two crests or two troughs coincide, there is constructive interference, marked by the red and blue dots. When a trough and a crest coincide, destructive interference occurs, marked by black dots. The phase difference is due to the path lengths traveled by the individual waves. Two identical waves travel two different path lengths to a point *P*.

(a) The difference in the path lengths is one wavelength, resulting in total constructive interference and a resulting amplitude equal to twice the original amplitude.

(b) The difference in the path lengths is less than one wavelength but greater than one half a wavelength, resulting in an amplitude greater than zero and less than twice the original amplitude.

(c) The difference in the path lengths is one half of a wavelength, resulting in total destructive interference and a resulting amplitude of zero.



The phase difference at each point is due to the different path lengths traveled by each wave. When the difference in the path lengths is an integer multiple of a wavelength,

$$\Delta r=|r2-r1|=n\lambda, where n=0,1,2,3,\ldots,$$

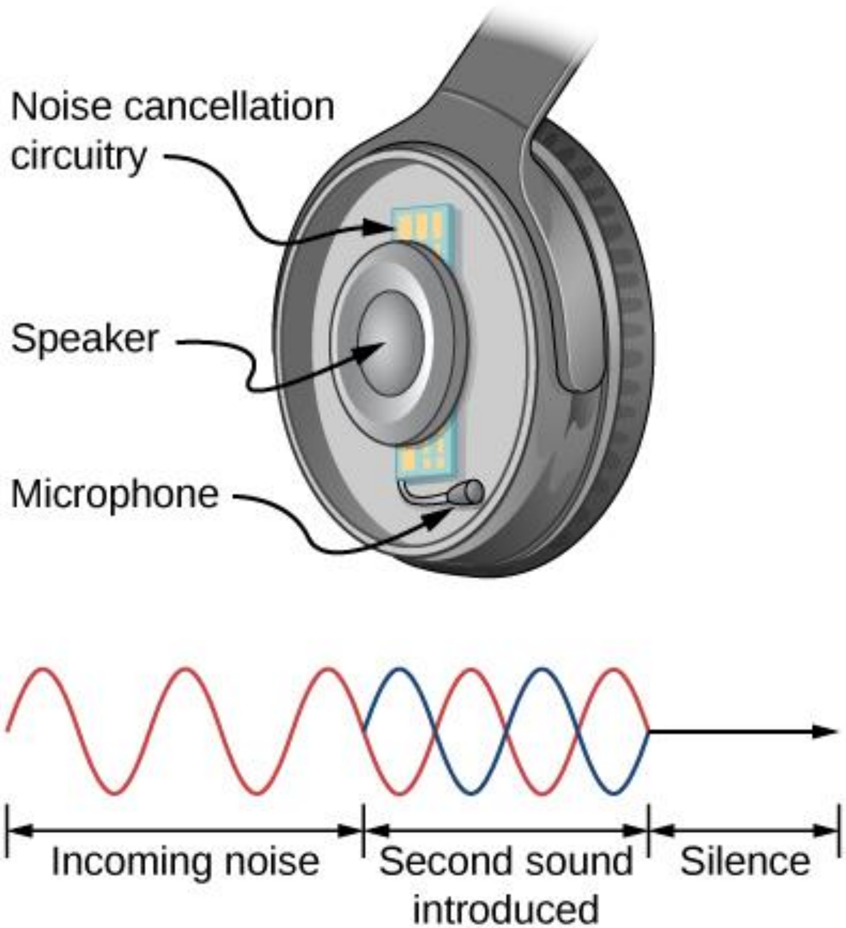
the waves are in phase and there is constructive interference. When the difference in path lengths is an odd multiple of a half wavelength,

$$\Delta r=|r2-r1|=n\lambda 2, where n=1,3,5,\ldots,$$

the waves are 180°(πrad) out of phase and the result is destructive interference. These points can be located with a sound-level intensity meter.

The lowest resonant frequency is called the **fundamental**, while all higher resonant frequencies are called **overtones**. The resonant frequencies that are integral multiples of the fundamental are collectively called **harmonics**. The fundamental is the first harmonic, the second harmonic is twice the frequency of the first harmonic, and so on. Some of these harmonics may not exist for a given scenario.

FIGURE 17.18



Headphones designed to cancel noise with destructive interference create a sound wave exactly opposite to the incoming sound. These headphones can be more effective than the simple passive attenuation used in most ear protection. Such headphones were used on the record-setting, around-the-world nonstop flight of the *Voyager* aircraft in 1986 to protect the pilots' hearing from engine noise.

Questions